

TEST CASES FILE

Overview

This Test Cases file defines structured scenarios used to validate the Smart Parking System. It ensures that all modules function correctly and meet requirements.

Test Categories

Type	Focus Area	Example
Unit Testing	Individual modules	Login validation
Integration Testing	Module interaction	UI + Database
System Testing	Full workflow	Complete booking process

Sample Test Cases

Unit Testing

Test ID	Module	Input	Expected Output
UT-01	Login	Valid credentials	Successful login
UT-02	Login	Invalid password	Error message
UT-03	Slot Check	Request availability	List of slots

Integration Testing

Test ID	Modules	Scenario	Expected Result
IT-01	UI + API	Fetch slots	Display updated slots
IT-02	Reservation + DB	Book slot	Slot status updated
IT-03	System + Notification	Booking confirmed	Alert sent

System-level test cases are also included to validate the complete workflow of the system, such as booking a parking slot, cancelling a reservation, and verifying real-time updates on the admin dashboard.

Each test case includes inputs, expected outputs, and actual results, allowing systematic validation of system behavior. These test cases ensure that the system is reliable, error-free, and aligned with user requirements before final deployment.

System Testing

Test ID	Scenario	Steps	Expected Result
ST-01	Full booking flow	Login → Book slot	Successful reservation
ST-02	Cancellation	Cancel booking	Slot becomes available
ST-03	Admin view	Open dashboard	Real-time data shown

Purpose of Testing

- Ensure system correctness
- Validate system reliability
- Detect logical errors early
- Improve system performance